A Longitudinal, Subject-Coherent Journal from Knesset Committee Protocols

Segmentation, Streaming Subject Linking, and Context-Conditioned Log Writing

Tomer Morad   Or Israeli   Yoel Marko

Advanced NLP (ANLP), July 2026

Abstract

We build a longitudinal journal over a stream of Israeli Knesset Finance Committee transcripts, coupling three tasks — topic segmentation, *streaming* cross-document subject linking, and update summarization — under an asymmetric cost model in which false merges are worse than false splits. Our central result is negative and, we argue, the most useful thing the project produced. On a manually annotated gold set of 523 segments, one fixed representation (`multilingual-e5-large`+ABTT) scores F_1=0.619 judging subject identity over *pairs*, and F_1=0.125 when the *same representation* makes the *same decision* online against a growing journal. The collapse is not caused by weak embeddings but by the structure of the streaming decision, where only 6.9% of arrivals are true repeats. We show what closes the gap — an accumulated-centroid entry representation lifts recall 32%→89%, and a distinctiveness *margin* rule raises F_1 0.103→0.190 — while every LLM verifier we tried stays $\leq$ 0.111. We further document two ways this benchmark silently rewards look-ahead, and report all headline numbers under a streaming-honest protocol.

1 Introduction

A Knesset committee returns to the same legislative matter repeatedly over months, so reconstructing the history of one matter means reading dozens of unrelated transcripts. We aim to produce that history automatically: an index in which each entry is one underlying matter carrying a dated log of its progress. This decomposes into (i) partitioning each protocol into subject spans, (ii) deciding online whether a new span continues an existing entry or opens a new one, and (iii) generating only the *incremental* update.

Asymmetric cost. Merging two distinct matters corrupts an entry permanently and silently; splitting one matter is visible and repairable. We therefore report every operating point twice: at best F_1, and at the *anti-duplication* point where the false-merge rate (FMR) is held $\leq$ 5%.

Research question. *Why does near-ceiling subject separability fail to produce usable streaming linking, and what closes the gap?* Our contributions: a gold standard over 211/213 protocols (§2); a quantification of the separability/decision gap (§4.1); an analysis of which levers close it (§4.2); evidence the LLM helps as an *extractor* not a *classifier* (§4.3); two look-ahead leaks and an honest protocol (§4.4); and a diagnosis of update summarization's recap-then-append failure (§5).

2 Data and Gold Standard

We use 213 Hebrew transcripts from the 25th Knesset Finance Committee. Protocol-level subject markers yield 151 canonical topics after fuzzy variant merging, but these describe the *meeting*, not the spans inside it, so we manually segmented 211/213 protocols:

Gold segments (subject spans)	523
Distinct subject labels	487
Recurring matters ($\geq$ 2 appearances)	22
Repeat events (decidable link decisions)	36
Base rate of repeats among arrivals	6.9%

We deliberately did *not* merge near-duplicate labels across protocols: under the asymmetric cost rule an unmerged duplicate is a cheap false split, whereas an incorrect merge would corrupt the gold itself. Recurrence counts are therefore conservative.

3 Segment Representation

Using only discussion *content* (no headers, agendas, attendees or speaker names), do segments of one matter cluster together? Off-the-shelf dense encoders are severely anisotropic: cross-topic cosine averages $\approx$ 0.95, leaving a same/different gap of 0.007–0.066, and under those conditions TF-IDF beat every dense model because the discriminative signal is lexical (bill numbers, entities). All-But-The-Top (ABTT) — centering and projecting out the top-k principal directions, which encode register rather than subject — reverses this.

Representation	ARI (agg.)	Pairwise AUC
e5, no ABTT (k=0)	0.124	0.857
e5 + ABTT (k=1)	**0.657**	**0.954**
TF-IDF	0.382	0.874
AlephBERT + ABTT (k=2)	0.265	0.854

Encoder choice matters, but de-anisotropization matters more. Removing a *single* direction moves e5 from ARI 0.124 to 0.657 — a larger effect than the spread across e5, TF-IDF and AlephBERT combined. We fix `e5+ABTT-k1` for all subsequent experiments.

4 Streaming Subject Linking

4.1 The separability/decision gap

We evaluate one representation under two protocols: **pairwise**, scoring all segment pairs (measuring *separability*, with oracle access to the full set), and **streaming**, where segments arrive chronologically and each is scored against the journal built so far — the actual task.

Protocol	AUC	F_1	P	R	FMR
Pairwise (oracle)	0.954	**0.619**	0.500	0.811	0.038
Streaming (real)	—	**0.125**	0.077	0.333	0.296

Central result. A representation separating same- from different-matter pairs at F_1 0.619 collapses to F_1 0.125 when the same decision is made online. **The bottleneck is the streaming decision structure, not representation quality.**

Why. Only 36 of 523 arrivals are repeats. Each is compared against a journal growing toward several hundred

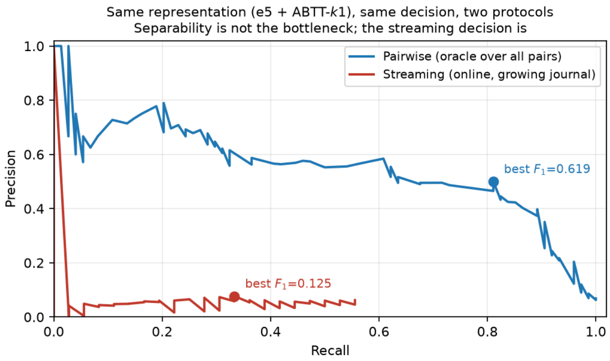

Figure 1: Precision–recall for one representation under both protocols. The oracle frontier sustains precision near 0.6; the streaming frontier never exceeds ≈ 0.08. The area between them is attributable to the decision setting, not the embedding.

entries, so with n candidates expected false merges scale as $n \cdot \mathrm{FPR}$ while available true positives stay at most one. A false-positive rate low enough to look excellent pairwise still floods the true matches — a base-rate effect, and precisely what makes anti-duplication hard. Notably, **retrieval was never the bottleneck**: similarity ranking places the correct entry in the top 10 for *all* 36 repeats (recall@1 0.694, @5 0.944, @10 1.000). The information is in the shortlist; the *decision* fails.

4.2 What closes the gap

From a baseline representing each entry by its frozen first span and linking above a cosine threshold, we vary one component at a time on frozen embeddings, CPU-only.

Method (streaming-honest)	F_1	R @ FMR$\leq$ 5%
Baseline: first span + threshold	0.103	2.8%
+ accumulated centroid	—	R 32%$\to$89%
+ distinctiveness margin	**0.190**	**8.6%**
Best LLM verifier (snippet)	0.111	$\approx$ 0%
LLM verifier, full timeline	0.068	0%

Represent the matter, not its first span. Replacing the frozen first span with the running centroid of an entry's members lifts recall 32% $\to$ 89%: the streaming setting continuously supplies evidence about what a matter looks like, and the baseline discards it.

Distinctiveness beats magnitude. Instead of "is the best candidate similar enough?" ($\cos \geq \theta$), the margin rule asks "is it *distinctly* more similar than the runner-up?" ($\cos_{(1)} - \cos_{(2)} \geq \theta$). This targets the real failure mode: the corpus contains families of near-identical bureaucratic templates, so high absolute similarity is uninformative while a large *gap* is not.

4.3 Extractor, not classifier

As a **verifier** deciding same/related/new from a candidate snippet, the LLM reached F_1 0.111; given the candidate's *full accumulated timeline* — the retrieve-then-verify design — it got *worse* (0.068), the longer context increasing its willingness to merge. As an **extractor** producing stable matter fingerprints (laws, programmes, entities) fused additively with the geometric score it helped: under a common batch-fit setting, geometry alone 0.200 vs. geometry+fingerprints 0.241 (`dictalm2`) and 0.201 (`qwen7b`).

Every LLM *classifier* variant scored at or below the CPU-only geometric method. The LLM's value was in *extracting stable identifiers*, leaving the decision to geometry.

4.4 Two look-ahead leaks

Streaming benchmarks make it easy to use future information unnoticed; we found two instances in our own pipeline. **(1) Batch-fit transforms.** ABTT and whitening estimate parameters from the embedding matrix; fitting once over the whole corpus uses segments that have not yet arrived. Refit honestly at each step from the observed prefix, the apparently strongest configuration collapses.

Transform	Batch (leaky)	Honest refit
Whitening (50-dim)	0.250	0.094
Centering	0.175	0.190
ABTT-k1	0.200	0.162

Whitening needs a 50-dimensional covariance estimate that does not exist early in the stream; cheap transforms needing few samples survive, expensive ones do not. **(2) Undefined cold-start margins.** With exactly one journal entry the runner-up is undefined; substituting a sentinel -1 makes the margin $\cos +1$, clearing any threshold, so the second segment of every run merges unconditionally. The correct fallback is the raw similarity magnitude.

Several configurations look like improvements under batch fitting and are partly measurement artifacts. **On streaming benchmarks the fitting protocol must be reported alongside the metric.**

5 Context-Conditioned Log Writing

Each update is conditioned on the running journal built from the model's *own* prior generations (faithfulness is chain-relative). Because predicted links are only $\approx$ 13% precise, evaluating through them would conflate two failures; we therefore evaluate on **gold chains** (22 matters, 58 entries), judged by a third model that is neither generator.

	`dictalm2`		`qwen2.5-7b`	
Prompt	Faith.	Nov.	Faith.	Nov.
Zero-shot	**80.7**	**18.1**	63.4	**10.1**
+ example (real)	71.1	15.0	63.1	8.3
+ example (fictional)	53.6	5.0	**69.2**	5.3

Faithfulness is acceptable — the Hebrew-native `dictalm2` is clearly more grounded than the stronger general-purpose model — but novelty is the failure. Rather than emitting a pure diff, models re-establish context around whatever new fact they add, so even the best examples reach only $\approx 50/100$; an explicit "do not repeat what is already logged" instruction does not prevent it.

Few-shot prompting made this monotonically worse for both models (18.1→15.0→5.0; 10.1→8.3→5.3). In the worst condition 92% of updates were near-verbatim copies of the example's answer: for a free-text *incremental-diff* task, examples are imitated as *content* rather than *style* — the opposite of the classification task in §4.3.

6 End-to-End Pipeline and Canonicalization

Composed end-to-end (embed → link → write) on all 523 gold segments under predicted growth, the anti-duplication operating point (θ=0.342) yields 493 entries of which 23 are multi-session, at 0.954 purity; the best-F_1 point (θ=0.156) yields 329 entries, 76 multi-session, at 0.658 purity. We treat this journal as a *demonstration artifact*, not an evaluation surface: at 0.13 linking precision an end-to-end score would conflate linking with summarization error, so each task is evaluated against gold inputs.

A parallel line of work in our group rewrites segments into neutral third-person Hebrew before embedding, targeting exactly the register confound of §4.2. On 19 protocols it sharpened intrinsic separation (substantive-segment AUC 0.947 → 0.986) but inverted downstream: F_1 0.114 → 0.095, with precision 0.073 → 0.154 and recall 0.261 → 0.069. It roughly doubles precision while collapsing recall — which under our asymmetric cost model is the *desirable* direction even though F_1 falls, meaning F_1 is the wrong instrument here. Two caveats prevent a stronger claim: the pseudo-labels come from the same LLM pass that produced the canonical text, and 29 repeat events is too thin to move F_1 stably.

7 Limitations and Conclusion

36 repeat events is few: recall at FMR$\leq$ 5% moves in 2.8-point increments, so large effects (representation, decision rule) are safe conclusions while fine rankings are not. Segmentation is gold rather than predicted, so an automatic segmenter would add its own error. All results come from one committee whose budget proceedings are unusually templated, and log-writing scores come from a 3B judge.

The intuitive failure mode — "Hebrew embeddings are not good enough" — is not what we found: a CPU-side representation separates same- from different-matter segments at AUC 0.954. What defeats the system is the streaming decision itself, where a 6.9% base rate against a growing candidate set means an excellent false-positive rate still produces more false merges than true ones. Consistent with that diagnosis, the levers that helped were structural rather than representational — represent the accumulated matter rather than its first appearance, and judge *distinctiveness* rather than *similarity* — while larger models in the deciding role did not help at all.

Reproducibility. The project is organised as numbered, self-contained steps (`steps/01..13`), each with its own `code/`, `outputs/` and `README.md`. Every number and figure traces to a committed JSON artifact under `steps/*/outputs/`; Fig. 1 is regenerated by `report/make_report_figs.py` with no hard-coded values.